

Panagiotis Kounatidis

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🌐 Website 🔗 LinkedIn 🐙 GitHub

About Me

I am PhD candidate in Systems Engineering at Cornell University specializing in optimization-based control, machine learning, reinforcement learning, and multi-agent systems. My work combines mathematical optimization and control theory with real-world implementation on robotic and autonomous systems. I have experience developing learning- and optimization-based algorithms, validating theoretical properties, and deploying control methods on physical multi-robot platforms and vehicle-related applications.

Technical Skills

Programming	C++, Python, MATLAB
Robotics	ROS 2, Gazebo, Nav2
Control	LQR/LQG, MPC, RL

Experience

Graduate Researcher — Information & Decision Science Lab, Cornell University **2024 – Present**
Ithaca, NY, USA

Research in optimization-based control, reinforcement learning, robotics, and autonomous systems under the supervision of Prof. Andreas Malikopoulos.

- Implemented a machine-learning-based algorithm for real-time multi-robot navigation with joint optimization of robot trajectories.
- Implemented a gradient-free stochastic optimization algorithm for linear systems and validated its theoretical robustness properties.
- Analyzed and implemented a new framework for optimal control problems with unknown dynamics based on dynamic programming and stochastic gradient descent.
- Analyzed and implemented a novel framework for safety-constrained continuous optimal control with unknown system dynamics and validated it on a cruise-control application using real wheeled robots.
- Operating seven LIMO ROS 2 robots and demonstrated their full functionality to high-profile laboratory visitors, including members of the Cornell University Board of Trustees.
- Co-hosting the DDLC seminars.

Lead Teaching Assistant — CEE 3040 – Uncertainty Analysis in Engineering, Cornell University **Fall 2025**
Ithaca, NY, USA

- Prepared instructional material, delivered lectures, discussion sessions, and office hours; proctored exams; and coordinated course logistics with the instructor, other TAs, and grader, for a class of ~ 100 students.

Lecturer & Grader — CEE 6080 – Optimal Control and Decision Theory, Cornell University **2025**
Ithaca, NY, USA

- Delivered lectures and graded student assignments.

Graduate Researcher — Institute of Automotive Technology, TUM **2022 – 2024**
Munich, Germany

Supervised by Dr. Simon Sagemster, Team Lead of TUM Autonomous Motorsport.

- Developed a vehicle model for systematic evaluation of vehicle motion controllers in simulation.
- Contributed to an extended version of the model that was integrated into the TUM Autonomous Motorsport software stack for controller evaluation.

IT Volunteer — European Championships 2022 **Aug. 2022**
Munich, Germany

Mechanical Designer & Manufacturer — Prom Racing Formula Student Team, NTUA **2019 – 2021**
Athens, Greece

- Designed and manufactured aerodynamic components of the Formula Student race car.
- Team achievement: 11 podium finishes during the 2020–2021 season, including 1st place overall at Formula Student Czech Republic.
- Team achievement: 6th place in static events at FS Online 2020.

Education

PhD Candidate in Systems Engineering , Cornell University <i>Ithaca, NY, USA</i>	2024 – Present
• Q-Exam: Passed • GPA: 4.140 / 4.300	
Diploma in Mechanical Engineering (Integrated B.S. & M.S.) , National Technical University of Athens (NTUA) <i>Athens, Greece</i>	2018 – 2023
• GPA: 8.82 / 10.00; ranked in the top 3.57%	
Erasmus Exchange Studies, School of Engineering & Design , Technical University of Munich (TUM) <i>Munich, Germany</i>	2022

Selected Publications

Combining Graph Attention Networks and Distributed Optimization for Multi-Robot Mixed-Integer Convex Programming V.-A. Le, P. Kounatidis, and A. A. Malikopoulos <i>IEEE 64th Conference on Decision and Control (CDC), 2025</i>	
Combined Learning and Control: A New Paradigm for Optimal Control with Unknown Dynamics P. Kounatidis, and A. A. Malikopoulos <i>American Control Conference (ACC), 2026</i>	
On the Robustness of Derivative-Free Methods for Linear Quadratic Regulator W. Li, P. Kounatidis, and A. A. Malikopoulos <i>IEEE Transactions on Automatic Control, 2026</i>	
Analyzing the Impact of Simulation Fidelity on the Evaluation of Autonomous Driving Motion Control S. Sagmeister, P. Kounatidis, S. Goblirsch and M. Lienkamp <i>IEEE Intelligent Vehicles Symposium (IV), 2024</i>	

Languages & Awards

Languages	English (fluent), German (fluent), French (elementary)
Awards	Cornell Systems Engineering Fellowship 2024–2025; IEEE CDC 2025 Student Travel Award; ACC 2026 Student Travel Award